

MSE 413 Final Project Report

Emotion Analysis Classification Based on Facial Coordinates

Name: Tom Dang
Student ID: 301548582

Name: Sam Drubetsky
Student ID: 301452017

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Introduction

The project uses several machine learning approaches to attempt to classify emotions based on facial landmark coordinates rather than using deep learning or neural network techniques. This attempts to quantify the impact of certain preconceived notions about what facial features actually represent what emotion. In addition, this project aims to evaluate the necessity of complicated neural network algorithms in this field, which are computationally expensive and time-consuming. This project aims to use simple machine learning techniques, such as KNN, random forests, and SVM, to reduce the learning time and improve the accuracy of classifier models.

Methodology

The dataset was sourced from Kaggle (msambare/fer2013), containing images labeled with seven different emotions: angry, disgust, fear, happy, sad, surprise, and neutral.

Face Detection and Landmark Extraction

Face Detection and Landmark Extraction is an OpenCV library and dlib's facial landmark detector that we used to streamline our data collection. As with this system, we could input images that had been preclassified and extract numerical data. Thereby allowing us to manipulate numerical data for classification instead of image data.

Next, from the landmark coordinates, we calculated some relevant facial features that we thought could be used to classify a person's emotion. The specific features used in this project are discussed later in the data exploration portion of this report.

Feature Selection and Dimensionality Reduction

To improve the model performance and interpretability, several dimensionality reduction techniques were applied:

- Highly correlated features (correlation > 0.8) were identified and removed
- StandardScaler was applied to normalize all features to minimize mean and unit variance.
- PCA techniques were used to visualize the data distribution in a three-dimensional space and better visualize if the data clusters based on facial geometry.
- For emotion-specific feature selection, `f_classif` was used to identify the most informative features for each emotion.

Classification Models

The study implemented and compared several machine learning models:

K-Nearest Neighbors (KNN)

- The optimal number of neighbors (k) was determined through cross-validation
- Multiple KNN strategies were tested, including:
 - A baseline model using all features
 - Emotion-specific models using the top 3 features for each emotion
 - A combined model using unique top features across all emotions

Support Vector Machine (SVM)

- Implemented with a Radial Basis Function (RBF) kernel
- Configured to generate probability estimates

Decision Tree

- Single decision tree model implemented as a baseline for comparison
- Poor Results led to using a Random Forest Classifier instead
 - Ensemble of 100 decision trees
- Feature importance was analyzed to identify the most predictive facial characteristics

Logistic Regression

- Second-degree fitting
- The maximum iteration parameter is set to 1000 to ensure convergence

Each model was evaluated using metrics, including accuracy, precision, recall, F1-score, and confusion matrices to provide a comprehensive understanding of classification performance.

Data exploration

The dataset used for emotion classification was sourced from Kaggle (msambare/fer2013), containing a selection of facial images labeled as one of seven emotions: angry, disgust, fear, happy, sad, surprise, and neutral. This dataset provides a broad selection of images for training and evaluating machine learning models with the goal of emotion recognition.



Figure 1: Sample Data for Emotion Classification. Source from Kaggle: msambare/fer2013

The dataset contained a substantial number of images across the seven emotion categories:

- Angry: 958 images
- Disgust: 111 images
- Fear: 1024 images
- Happy: 1774 images
- Sad: 1247 images
- Surprise: 831 images
- Neutral: 1233 images

This distribution reveals a notable imbalance, with the "disgust" category being quite underrepresented compared to other emotions. The "happy" category contains the highest number of samples, with approximately 16 times more images than the "disgust" category. This inequality will be a notable issue explored later as it results in a bias for most models.

Using OpenCV and dlib's facial landmark detector, we processed the images and extracted 68 facial landmark coordinates from each image. These coordinates allow for the identification of key facial points, including the contours of the eyes, eyebrows, nose, lips, and jawline. From these coordinate values, several features that were considered relevant were derived:

- Mouth measurements (width, height, aspect ratio)
- Eye measurements (width, height, aspect ratio for both eyes)
- Eyebrow positions relative to eyes
- Distances between various facial points

After the feature extraction process, our training set consisted of 3,966 samples by 156 features, while the testing set contained 992 samples. This drop in numbers from the original dataset size means that the facial landmark detection algorithm was unable to process all images correctly and discarded several.

Feature Analysis

To understand the effect of some notable features, the distribution across different emotions were plotted.

The mouth aspect ratio (height to width ratio) shows distinct patterns across emotions. As expected, emotions like "surprise" typically exhibit more exaggerated mouth aspect ratios on account of having a more open mouth. In contrast, emotions like "angry" and "neutral" tend to show lower mouth aspect ratios, reflecting how people typically parse their lipase when angry or simply keep their mouth closed in neutral scenarios.

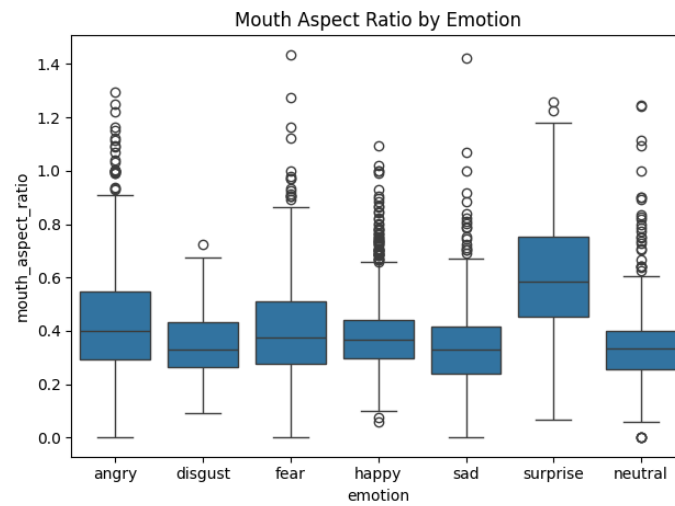


Figure 2.1: Data Distribution of Mouth Aspect Ratio Across 7 Different Emotions

The average eye aspect ratio (height to width ratio of eyes) demonstrates further emotion-specific patterns. "Surprise" and "fear" emotions generally display higher eye aspect ratios, corresponding to wider-open eyes. Whilst disgust has the smallest eye ratio, perhaps indicative of people grimacing.

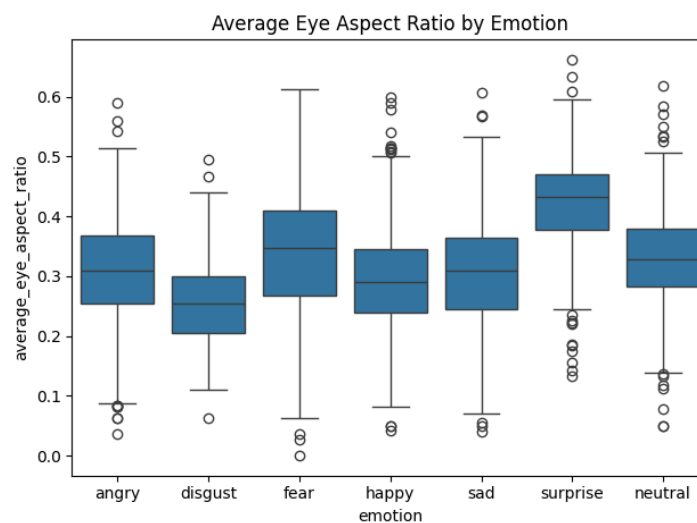


Figure 2.2: Data Distribution of Average Eye Aspect Ratio Across 7 Different Emotions

The average distance between eyebrows and eyes relative to eye height reveals interesting patterns across emotions. "Angry" emotions often display smaller eyebrow-eye distances as the eyebrows tend to be lowered. Also, "surprised" surprisingly has the lowest eyebrow distance despite our hypothesis that raised eyebrows are a symbol of surprise. Whilst happy and sad show the highest eyebrow ratio.

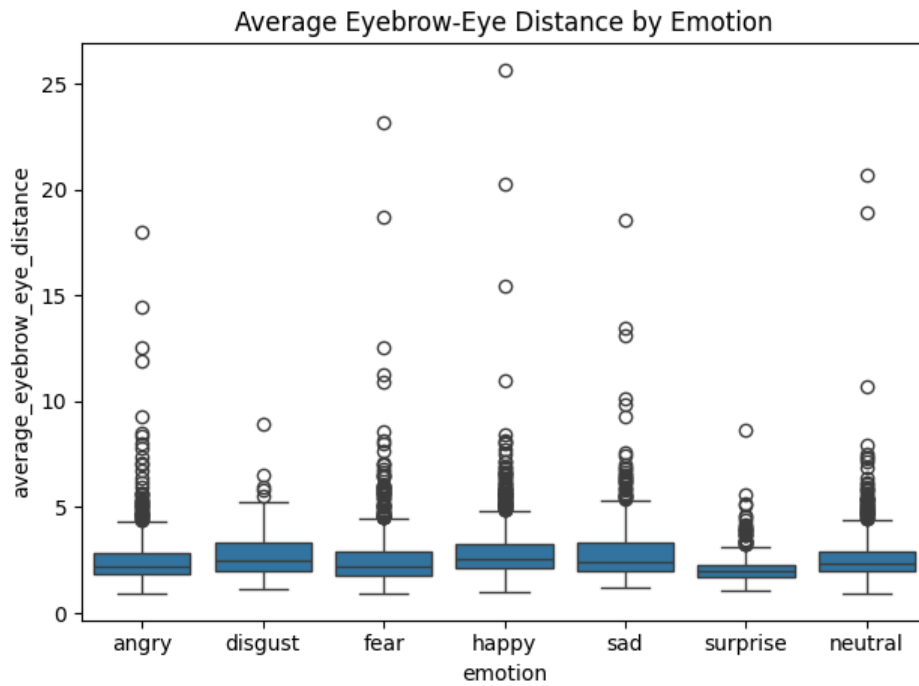


Figure 2.3: Data Distribution of Average Eyebrow Distance Across 7 Different Emotions

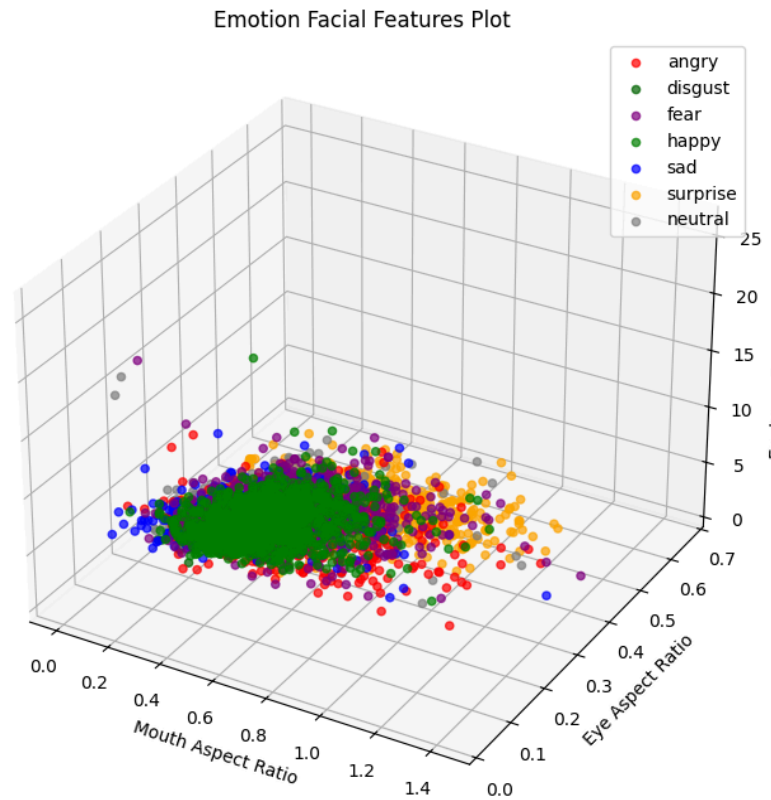


Figure 3: Data Distribution of Emotions Along 3 Suspected Features

Dimensionality Reduction and Feature Importance

After calculating correlation matrices for our features, we identified and removed redundant features that correlate greater than .8. This is to reduce redundancy in the dataset and allow more efficiency processing in later stages of the project. Standard scaling was employed to normalize all features, minimizing the mean and achieving unit variance to ensure fair comparison between features of different scales.

```
Columns dropped: ['mouth_aspect_ratio', 'right_eye_height', 'average_eye_height', 'left_eye_aspect_ratio',  
Columns in training set: Index(['mouth_width', 'mouth_height', 'left_eye_width', 'right_eye_width',  
                                'left_eye_height', 'left_eyebrow_eye_distance',  
                                'average_eyebrow_eye_distance', 'x_0', 'y_0', 'x_17', 'y_48'],
```

Figure 4: Dropped and Undropped Columns After Normalization and Scaling

Principal Component Analysis (PCA) was employed to visualize the data distribution in a three-dimensional space. The PCA visualization revealed a large cluster of emotions that was difficult to piece together from the plot.

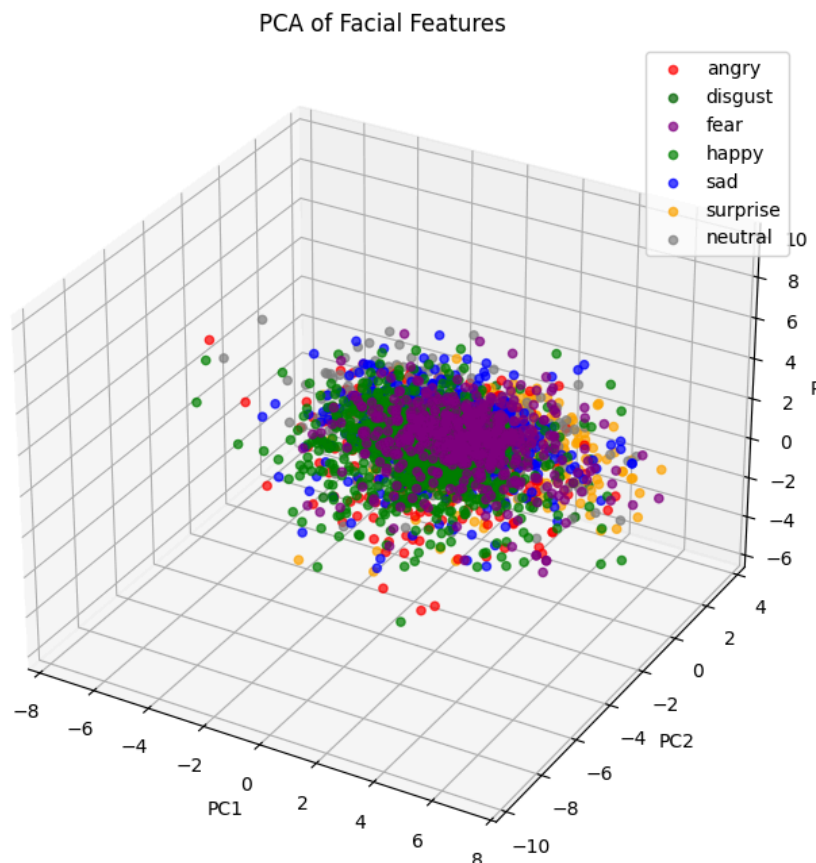


Figure 5: PCA Distribution of Emotions

Results and Discussions

KNN Classification

Several KNN approaches were tested:

- The baseline model using all features achieved moderate results
- Emotion-specific models using the top 3 features for each emotion showed varying effectiveness
- A combined model using unique top features across all emotions provided the best KNN results

The first KNN model running on all features found an optimal K value of 10 with an accuracy of 46%. When employing 7 clusters (an equal amount to the input emotions) the KNN model had an accuracy of 43.5%.

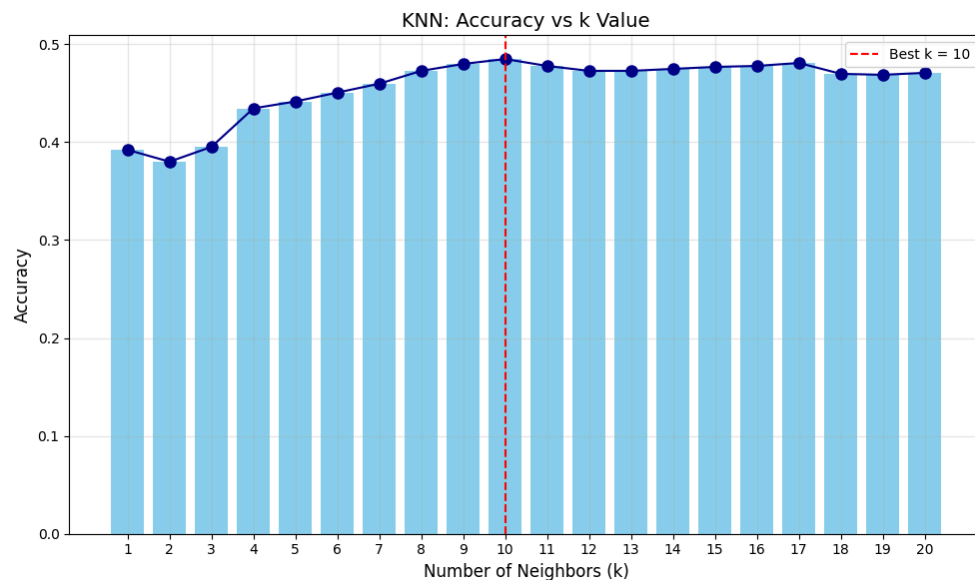


Figure 6: KNN Accuracy Corresponding to the K Neighbors for All Features

When running KNN models with different features to isolate for certain emotions, the difference between accuracy and F1 score became increasingly notable as disgust was best classified with 2 clusters yielding an accuracy of 98% but an F1 score of 0. Meaning it was completely missattributed and the sheer lack of datapoints resulted in poor classification. The only meaningful data is that happiness can be classified with an f1 score of 61%, with mouth width, left eye height and y_48 (coinciding with right eye position)

```

Top 3 features for 'angry':
- mouth_width
- left_eyebrow_eye_distance
- y_0
Best k for 'angry': 8. Accuracy: 0.8518. F1: 0.0134

Top 3 features for 'disgust':
- right_eye_width
- left_eye_height
- y_48
Best k for 'disgust': 2. Accuracy: 0.9819. F1: 0.0000

Top 3 features for 'fear':
- left_eye_width
- left_eye_height
- y_48
Best k for 'fear': 8. Accuracy: 0.8690. F1: 0.0000

Top 3 features for 'happy':
- mouth_width
- left_eye_height
- y_48
Best k for 'happy': 10. Accuracy: 0.8125. F1: 0.6141

Top 3 features for 'sad':
- mouth_width
- mouth_height
- left_eye_height
Best k for 'sad': 8. Accuracy: 0.8750. F1: 0.0159

Top 3 features for 'surprise':
- mouth_height
- left_eye_height
- left_eyebrow_eye_distance
Best k for 'surprise': 9. Accuracy: 0.8871. F1: 0.4167

Top 3 features for 'neutral':
- mouth_width
- mouth_height
- y_48
Best k for 'neutral': 2. Accuracy: 0.8054. F1: 0.1422

```

Figure 7: Optimal Feature Selection for KNN Analysis of Individual Emotion

Next a model was made from the important features in the above KNN models in hopes of using features only relevant to each emotion. Nevertheless this model underperformed the original.

```

===== PERFORMANCE COMPARISON =====

Baseline (All Features):
Accuracy: 0.4849
angry: F1=0.2845
disgust: F1=0.0909
fear: F1=0.2735
happy: F1=0.7131
sad: F1=0.1828
surprise: F1=0.6325
neutral: F1=0.4635

Ensemble of Emotion-Specific Models (3 features each):
Accuracy: 0.3962
angry: F1=0.2338
disgust: F1=0.0455
fear: F1=0.1872
happy: F1=0.6471
sad: F1=0.0722
surprise: F1=0.5242
neutral: F1=0.3476

Combined Top Features Model (8 features):
Accuracy: 0.4657
angry: F1=0.3077
disgust: F1=0.0000
fear: F1=0.0905
happy: F1=0.6968
sad: F1=0.2128
surprise: F1=0.5796
neutral: F1=0.4593

```

Figure 8: KNN Accuracy Comparison for Different Models

The confusion matrix for our best KNN model reveals increased accuracy towards emotions that had larger input data. Neutral had the highest accuracy with 74 correct classifications and 31 misses, whilst disgust had 1 correct classification and missclassifications. It is interesting to pay attention to how many points fall within each category, as happiness has by far the most classifications, followed by sad aligning with the amount of training data input.

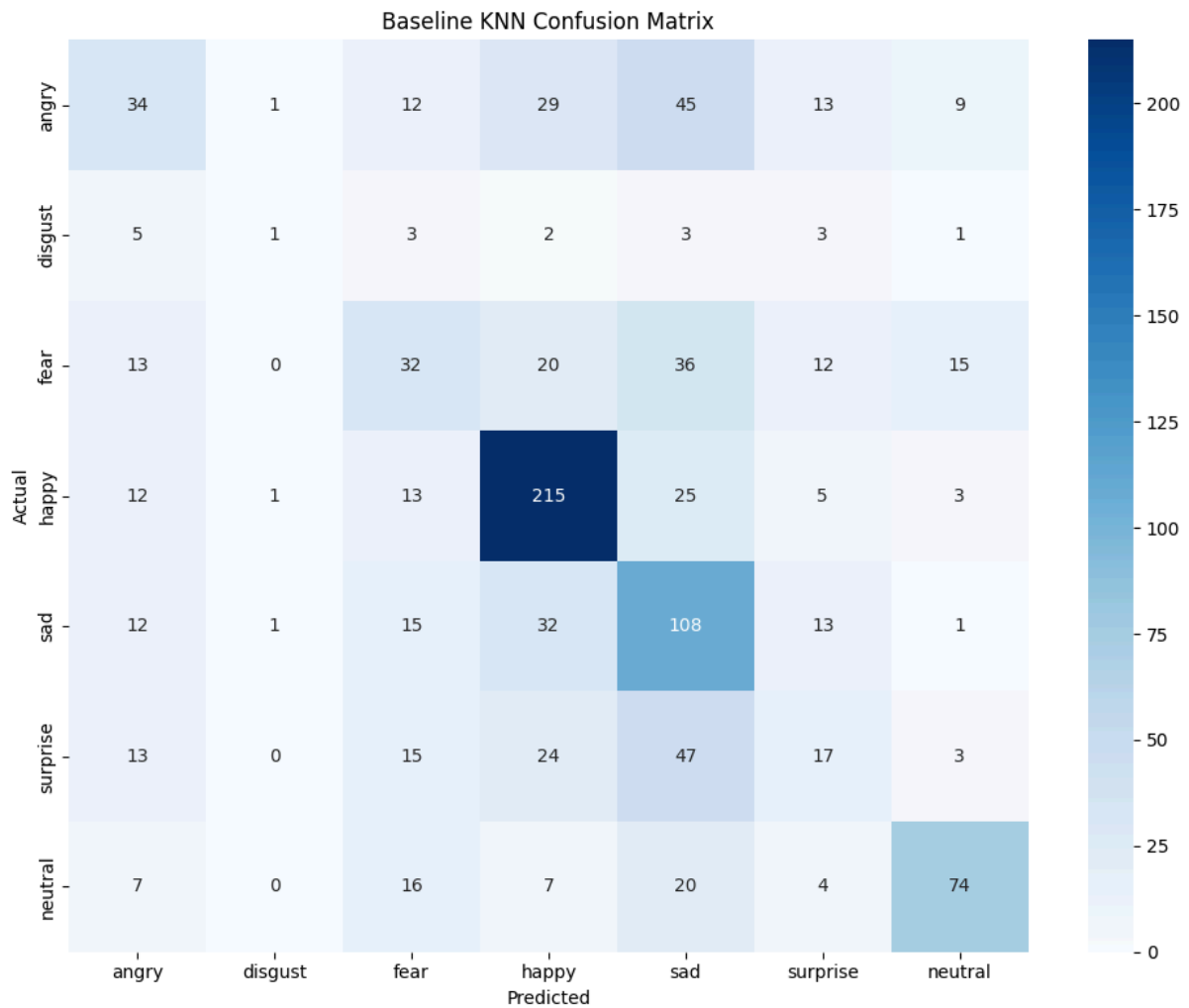


Figure 9: Best KNN Model Confusion Matrix

SVM

The Support Vector Machine model with an RBF kernel achieved the second highest overall accuracy at 56%, outperforming other most other classifiers. This suggests that the non-linear decision boundaries created by the SVM are better at capturing the relationships between facial landmarks and emotional states.

Table 1: SVM Model Results

```

--- Support Vector Machine (SVM) ---
SVM Accuracy: 0.4929

SVM Classification Report:

```

	precision	recall	f1-score	support
angry	0.47	0.27	0.34	143
disgust	0.00	0.00	0.00	18
fear	0.27	0.10	0.15	128
happy	0.68	0.79	0.73	274
neutral	0.38	0.65	0.48	182
sad	0.24	0.18	0.21	119
surprise	0.59	0.63	0.61	128
accuracy			0.49	992
macro avg	0.37	0.38	0.36	992
weighted avg	0.46	0.49	0.46	992

While SVM performed relatively better, the confusion matrix reveals persistent challenges in differentiating between certain emotion pairs. This model completely failed to acknowledge disgust due to its low sample size and just did not classify any image as disgust.

The SVM's superior performance can be attributed to its ability to find optimal separation boundaries in high-dimensional spaces and its better resilience to the curse of dimensionality which is crucial when classifying things with so many features.

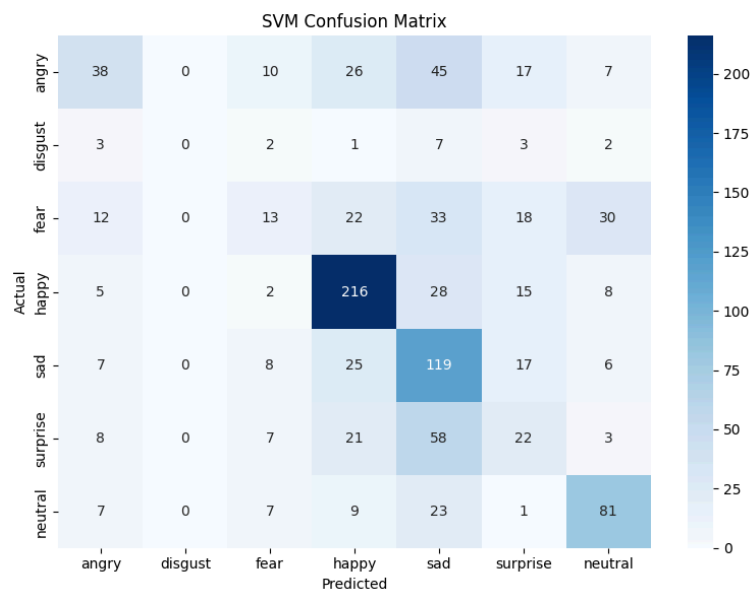


Figure 10: SVM Model Confusion Matrix

Random Forrest

The decision tree classifier was implemented as a baseline model to establish a performance benchmark before moving to more complex ensemble methods. According to the results presented in Table 1 (Decision Tree Model Results), the single decision tree demonstrated the worst performance out of all experimented models. The model achieved an accuracy of only 37%.

Table 2: Decision Tree Model Results

```
--- Decision Tree ---
Decision Tree Accuracy: 0.3700

Decision Tree Classification Report:
              precision    recall  f1-score   support

   angry         0.27         0.22         0.24         143
   disgust        0.09         0.11         0.10          18
    fear         0.21         0.21         0.21         128
   happy         0.58         0.58         0.58         274
   neutral        0.31         0.32         0.31         182
    sad          0.19         0.21         0.20         119
   surprise       0.51         0.50         0.51         128

 accuracy                   0.37         992
 macro avg         0.31         0.31         0.31         992
 weighted avg      0.37         0.37         0.37         992
```

However, the Random Forest classifier, with a selection of 100 decision trees, performed far better than the and had the highest accuracy amongst all the models. The feature importance analysis from the Random Forest revealed valuable insights into which facial landmarks contribute most significantly to emotion classification:

Table 3: Random Forest Tree Model Results

```
--- Random Forest ---
Random Forest Accuracy: 0.4990

Random Forest Classification Report:
              precision    recall  f1-score   support

   angry         0.42         0.24         0.31         143
   disgust        0.50         0.06         0.10          18
    fear         0.40         0.19         0.26         128
   happy         0.64         0.81         0.71         274
   neutral        0.39         0.56         0.46         182
    sad          0.29         0.23         0.25         119
   surprise       0.59         0.66         0.62         128

 accuracy                   0.50         992
 macro avg         0.46         0.39         0.39         992
 weighted avg      0.48         0.50         0.47         992
```

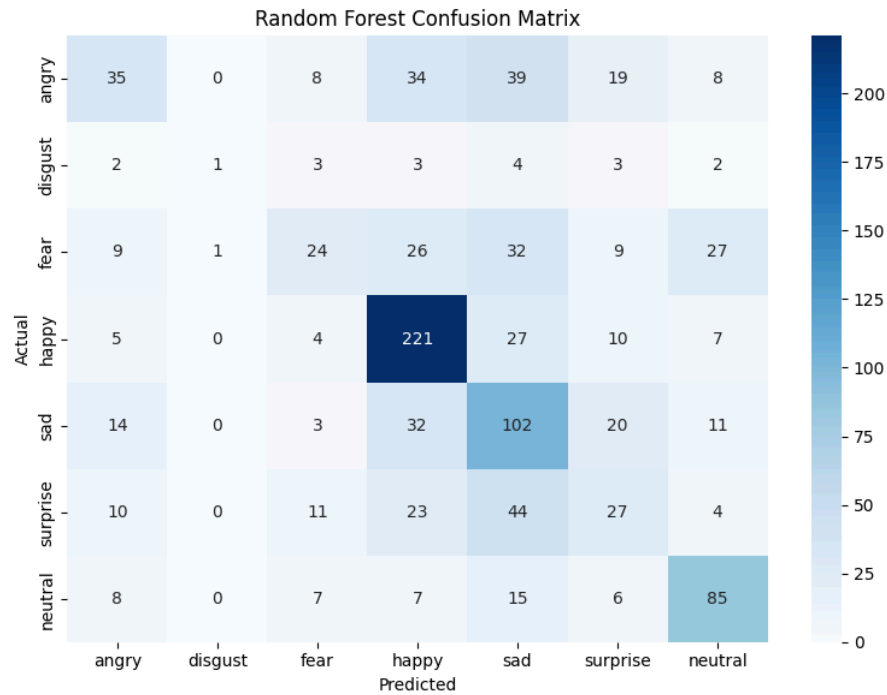


Figure 11: Random Forest Confusion Matrix

The top features identified by the Random Forest classifier were primarily related to:

- Mouth measurements (aspect ratio and positioning)
- Eye openness measurements
- Eyebrow positioning relative to eyes

These findings align with our initial predictions and separate KNN models which faboered moth aspect ratio, eye aspect ratio, and eyebrow distance across different emotions.

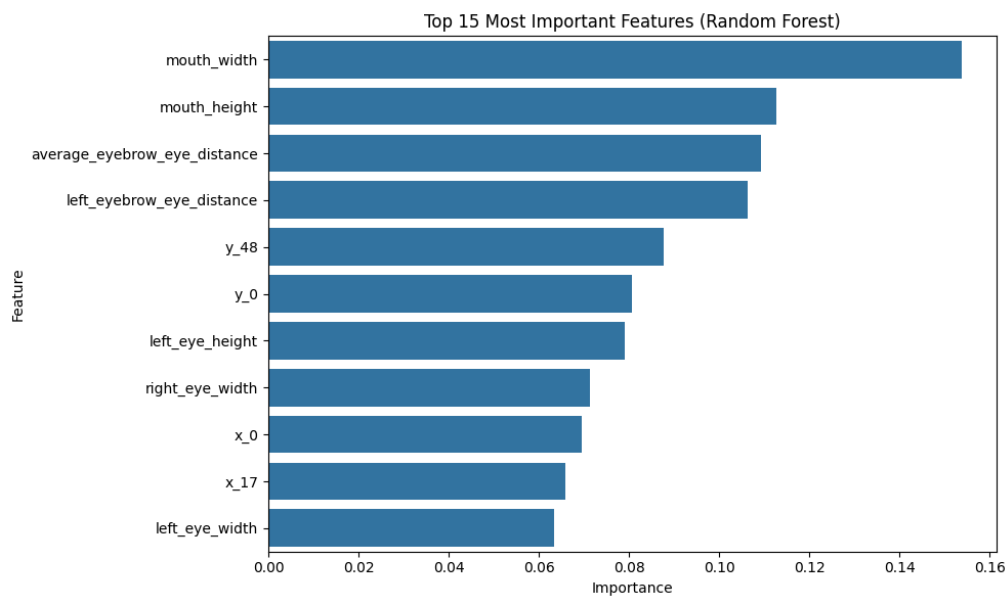


Figure 12: Top 15 Features According to Random Forest Classifier

Logistic Regression

The Logistic Regression model with second-degree fitting achieved the second lowest accuracy among our tested models, just above the single decision tree. As noted in the model results talbe, this model exhibited a high false positive rate, indicating a tendency to overpredict certain emotion classes. This suggests that while the model captures some patterns in the data, it struggles with the clear separation of emotion categories.

Table 4: Logistic Regression Model Results

```
--- Logistic Regression with Polynomial Features ---
Logistic Regression with Polynomial Features Accuracy: 0.3942

Classification Report:
              precision    recall  f1-score   support

   angry         0.50         0.01         0.01         143
   disgust        0.00         0.00         0.00          18
    fear         0.00         0.00         0.00         128
   happy         0.48         0.76         0.59         274
   neutral        0.28         0.61         0.38         182
    sad          0.32         0.06         0.10         119
   surprise       0.47         0.51         0.49         128

   accuracy                   0.39         992
  macro avg         0.29         0.28         0.22         992
 weighted avg         0.35         0.39         0.31         992
```

Model Comparison

The evaluation of models for this project reveals significant performance differences, with Random Forest achieving the highest accuracy (49.9%), followed closely by SVM (49.3%), KNN (48.5%), and then Logistic Regression (39.4%) and Decision Tree (37.0%).

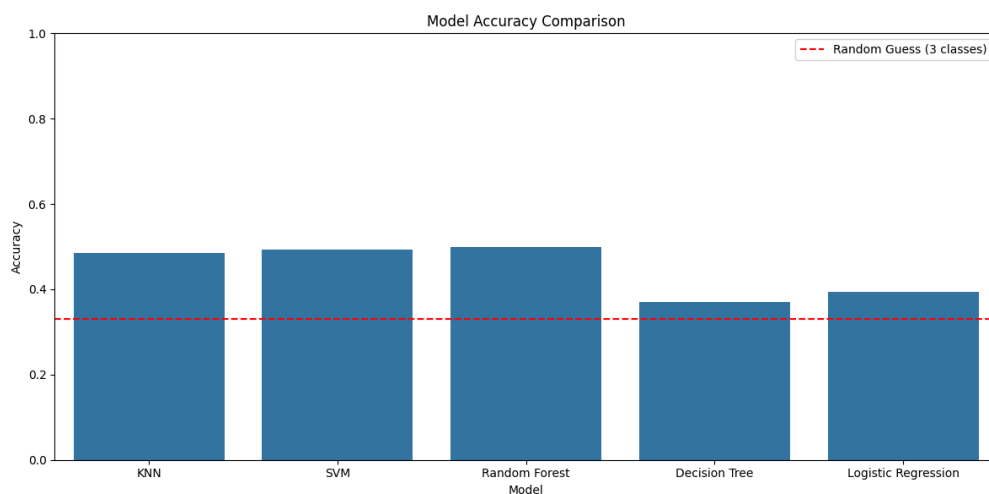


Figure 13: Comparing Accuracy of all Models

The performance of Random Forest and SVM indicates that non-linear approaches are better suited for capturing the relationships between facial geometries and emotion classification. Random Forest's method effectively reduced overfitting, whilst SVM's RBF created the non-linear boundaries that were mostly effective for this task.

An interesting observation from the confusion matrices revealed consistent misclassification patterns, with "fear" being classified as "neutral" across multiple models. This suggests similarities in facial expressions between the two despite the intuitive difference. Furthermore, all models really struggled with the "disgust" category due to severe class imbalance (only 111 samples compared to 1774 for "happy").

The main challenges with this dataset and approach are as follows:

- Class Imbalances: The lack of images of "disgust" (111 samples) compared to "happy" (1774 samples) created a bias in all models toward predicting the majority classes.
- Feature Overlap: Many emotions share similar facial configurations, making it difficult to establish clear decision boundaries based solely on geometric features.
 - Perhaps choosing 3 distinct emotions such as happy, sad and angry would allow for a more distinct difference in results
- Dimensionality Challenges: Despite our attempts at feature selection and dimensionality reduction efforts, the high-dimensional nature of the problem (156 features) compared to the more limited sample size of 3,966, has led to overfitting in some models.

The PCA visualization highlighted some of these challenges, showing a lot of overlap between emotion clusters making them visually indistinguishable. Suggesting that linear separation of emotions based on facial landmarks alone is nearly impossible, explaining why models like SVM performed better.

Conclusion

This project evaluated traditional machine learning approaches for emotion classification using facial landmark coordinates as an alternative to computationally expensive deep learning methods. Our analysis revealed that Random Forest (49.9% accuracy) and SVM (49.3% accuracy) classifiers performed best among the tested models, demonstrating that geometric facial features do contain meaningful emotional information. Though yielding lower accuracy than what would be considered "reliable" for practical applications, it is important to note that with seven possible emotion classes, these results are substantially better than random guessing (14.3%).

Several challenges impeded performance, including significant class imbalance in the dataset (with "disgust" having only 111 samples compared to "happy" with 1774) and feature ambiguity between emotions that share similar facial configurations. Particularly interesting was the that "fear" was frequently misclassified as "neutral" across multiple models, suggesting these emotions may share similar geometric facial characteristics.

Feature importance analysis confirmed our hypothesis that mouth measurements, eye openness metrics, and eyebrow positioning were the critical features for emotion classification. The superior performance of non-linear models like Random Forest and SVM compared to KNN (48.5%), Logistic Regression (39.4%), and Decision Tree (37.0%) suggests that the relationship between facial geometry and emotions requires complex algorithms that simple models cannot capture.

While traditional machine learning methods can identify some aspects of emotional expression through facial landmarks, they lack the accuracy needed for reliable emotion recognition systems. Furthermore, this approach does not end up more computationally efficient, as the data provided for this model was filtered through a computer vision library. Meaning there is zero to very little saving in computational power, as classifying points on a face would be incredibly time-consuming and ultimately redundant compared to any ANN computer vision approaches.

References

Data source: [1]"Human Face Emotions," [www.kaggle.com](https://www.kaggle.com/datasets/msambare/fer2013).
<https://www.kaggle.com/datasets/msambare/fer2013>